

Lecture 30:

Moment Generating Function

$$M_X(s) = E[e^{Xs}] \quad s \in \mathbb{R}.$$

For discrete R.V. X

$$M_X(s) = \sum_{x \in \mathcal{X}_0} e^{xs} P_X(x)$$

For continuous R.V. X

$$M_X(s) = \int_{-\infty}^{\infty} e^{xs} f_X(x) dx.$$

May not be well defined for all $s \in \mathbb{R}$.

Examples :

① $X \sim \text{Bernoulli}(p)$

$$\begin{aligned} M_X(s) &= p \times e^{s \times 1} + (1-p) e^{s \times 0} \\ &= p e^s + (1-p). \end{aligned}$$

② $X \sim \text{Exponential}(\lambda)$

$$M_X(s) = \int_0^{\infty} \lambda e^{-\lambda x} e^{sx} dx.$$

$$= \lambda \int_0^{\infty} e^{-x(\lambda - s)} dx$$

$$= \begin{cases} \frac{\lambda}{\lambda - s} & \lambda - s > 0. \\ \text{not well defined} & \text{otherwise} \end{cases}$$

$$\textcircled{3} \quad X \sim N(0, 1).$$

$$M_X(s) = \int_{-\infty}^{\infty} \frac{e^{sx} e^{-x^2/2}}{\sqrt{2\pi}} dx.$$

$$= \int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}(x^2 - 2sx)}}{\sqrt{2\pi}} dx$$

$$= e^{s^2/2} \int_{-\infty}^{\infty} \frac{e^{-\frac{1}{2}(x-s)^2}}{\sqrt{2\pi}} dx$$

$$= e^{s^2/2}.$$

④ $X \sim \text{Poisson}(\lambda)$.

$$\begin{aligned} M_X(s) &= E[e^{Xs}] = \sum_{x=0}^{\infty} e^{xs} \left[\frac{\lambda^x e^{-\lambda}}{x!} \right] \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(e^s \lambda)^x}{x!} \\ &= e^{-\lambda} e^{e^s \lambda} = e^{\lambda(e^s - 1)}. \end{aligned}$$

Properties of this transform.

① $Y = aX + b$, $M_Y(s)$ ^{determine} from $M_X(s)$.

$$M_Y(s) = E[e^{Ys}] = E[e^{(ax+b)s}]$$

$$= e^{bs} E[e^{as}]$$

$$= e^{bs} M_X(as)$$

② $Y = X_1 + X_2$ X_1, X_2 are independence

$$M_Y(s) = M_{X_1}(s) M_{X_2}(s).$$

In general for $Y = X_1 + X_2 + \dots + X_n$

$$M_Y(s) = \prod_{i=1}^n M_{X_i}(s) \quad X_1, \dots, X_n \text{ are independence.}$$

Example:

① $X_1 \sim N(0, 1), \quad X_2 \sim N(0, 1).$

$$Y = X_1 + X_2 \sim N(0, 2).$$

$$M_{X_1}(s) = e^{s^2/2} = M_{X_2}(s)$$

$$M_Y(s) = e^{s^2/2} e^{s^2/2} = e^{s^2}.$$

Suppose $Z \sim N(\mu, \sigma^2)$.

$$M_Z(s) = ?$$

Suppose N is standard normal R.V.

$$N \sim N(0, 1)$$

$$Z = \sigma N + \mu.$$

$$E[Z] = \sigma \underbrace{E[N]}_0 + \mu = \mu$$

$$\text{Var}(Z) = \sigma^2 E[N^2] = \sigma^2.$$

$$M_Z(s) = E[e^{zs}]$$

$$= E[e^{(\sigma N + \mu)s}]$$

$$= e^{\mu s} E[e^{N\sigma s}]$$

$$M_N(s) = \frac{s^2}{2}$$

$$= e^{\mu s} M_N(\sigma s)$$

$$= e^{\mu s} e^{\sigma^2 s^2 / 2}$$

$$\text{For } \mu=0, \sigma^2=2, M_Z(s) = e^{s^2}.$$

$$\therefore Y = X_1 + X_2 \sim N(0, 2).$$

②

$$Y = X_1 + \dots + X_n \quad X_i \text{'s are iid.}$$

$\sim \text{Bernoulli}(p)$.

$$\begin{aligned} M_Y(s) &= \prod_{i=1}^n (M_{X_i}(s)) \\ &= (M_{X_1}(s))^n \\ &= ((1-p) + pe^s)^n. \end{aligned}$$

③

$$Y = X_1 + X_2.$$

$X_1 \sim \text{Poisson}(\lambda_1)$.

$X_2 \sim \text{Poisson}(\lambda_2)$.

$$M_Y(s) = M_{X_1}(s) M_{X_2}(s).$$

$$= e^{\lambda_1(e^s - 1)} \cdot e^{\lambda_2(e^s - 1)}$$

$$= e^{(\lambda_1 + \lambda_2)(e^s - 1)} \Rightarrow Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

Inversion Property

$$\textcircled{3} M_X(s) = E[e^{sx}] = E\left[1 + sx + \frac{(sx)^2}{2} + \frac{(sx)^3}{3!} + \dots\right]$$

$$\frac{\partial M_X(s)}{\partial s} = E\left[\frac{\partial e^{sx}}{\partial s}\right] = E\left[x e^{sx}\right]$$

$$= E\left[x + \frac{2sx^2}{2} + \frac{3s^2x^3}{3!} + \dots\right]$$

$$\left.\frac{\partial M_X(s)}{\partial s}\right|_{s=0} = E[x e^{sx}]_{s=0}$$

$$= E[x]$$

$$M_X(0) = E[e^{0 \cdot x}] = 1$$

$$\frac{\partial^2 M_X(s)}{\partial s^2} = E \left[\frac{\partial^2 X e^{sX}}{\partial s^2} \right]$$

$$= E [X^2 e^{sX}]$$

$$\Rightarrow \left. \frac{\partial^2 M_X(s)}{\partial s^2} \right|_{s=0} = E[X^2]$$

⋮

$$\frac{\partial^m M_X(s)}{\partial s^m} = E [X^m e^{sX}]$$

$$\left. \frac{\partial^m M_X(s)}{\partial s^m} \right|_{s=0} = E [X^m]$$

Example :

$$X \sim \text{Exp}(\lambda).$$

$$M_X(s) = \frac{\lambda}{\lambda - s} \quad s < \lambda$$

$$M_X(0) = 1.$$

$$\begin{aligned} \frac{\partial M_X(s)}{\partial s} &= \lambda \left(\frac{-1}{\lambda - s} \right)^2 \times (-1) \\ &= \frac{\lambda}{(\lambda - s)^2}. \end{aligned}$$

$$E[X] = \left. \frac{\partial M_X(s)}{\partial s} \right|_{s=0} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

$$\frac{\partial^2 M_X(s)}{\partial s^2} = \frac{2\lambda}{(\lambda - s)^3}.$$

$$E[x^2] = \frac{\partial^2 M_x(s)}{\partial s^2} \bigg|_{s=0}.$$

$$= \frac{2}{\lambda^2}.$$

$$E[x^m] = \frac{m!}{\lambda^m}.$$

③ Inversion property: The transform $M_X(s)$ completely determines the probability law of R.V. X . In particular if $M_X(s) = M_Y(s) \forall s$ then R.V.s X, Y have same probability law

④ $Y = X_1 + X_2 + \dots + X_N$ (Poisson splitting)
 N is independent of $X_1, X_2, \dots$
 $X_1, X_2, \dots$ are i.i.d.

$M_Y(s) = E[e^{sY}]$
 $= E[E[e^{sY} | N]]$

$\sum_n P_{Y|N}(y|n) P_N(n) = P_Y(y)$

$\xrightarrow{N} \frac{p}{1-p}$

$X_i \sim \text{Bernoulli}(p)$.
 # of packets scheduled in line 1 = $X_1 + X_2 + \dots + X_N$
 $X_1, X_2, \dots, N$ are

$$= \sum_n P(N=n) E[e^{Y_s} | N=n] \quad \left. \begin{array}{l} \text{independent} \end{array} \right\}$$

$$E[e^{Y_s} | N=n] = E[e^{(X_1 + X_2 + \dots + X_N)s} | N=n]$$

$$= E[e^{(X_1 + X_2 + \dots + X_n)s} | N=n]$$

$$= E\left[\prod_{i=1}^n e^{X_i s} \mid N=n\right]$$

X_i 's independent
of N $\leftarrow$

$$= E\left[\prod_{i=1}^n e^{X_i s}\right]$$

X_i 's are independent $\Rightarrow$

$$= \prod_{i=1}^n E[e^{X_i s}] = \prod_{i=1}^n M_{X_i}(s)$$

X_i 's are identical $\rightarrow = (M_{X_1}(s))^n$.

$$M_Y(s) = \sum_n P(N=n) E[e^{Xs} | N=n].$$

Given $M_N(s)$ & $M_{X_1}(s)$ at all s
find $M_Y(s)$

$$= \sum_n P(N=n) M_{X_1}(s)^n.$$

$$= E[(M_{X_1}(s))^N].$$

$$= E[e^{\log(M_{X_1}(s))N}].$$

$$= M_N(\log(M_{X_1}(s))).$$

Example:

$N \sim \text{Poisson}(\lambda)$.

$X_1 \sim \text{Bernoulli}(p)$.

X_i 's are iid
 $X_1, \dots, X_N$ are independent

$$Y = X_1 + X_2 + \dots + X_N$$

$$M_Y(s) = M_N(\log M_{X_1}(s)).$$

$$M_{X_1}(s) = (1-p) + p e^s, \quad M_N(s) = e^{\lambda(e^s - 1)}$$

$$\begin{aligned} M_Y(s) &= M_N(\log((1-p) + p e^s)) \\ &= e^{\lambda [e^{\log((1-p) + p e^s)} - 1]} \\ &= e^{\lambda [(1-p) + p e^s - 1]} \end{aligned}$$

$$= e^{\lambda [e^s - 1]}.$$

↓

This is MGF of Poisson (λ).

$$Y \sim \text{Poisson}(\lambda).$$

by inversion property.

⑤

Mixture distributions.

$$f_Y(y) = \sum_{i=1}^n p_i f_{X_i}(y).$$

Example:

$$\begin{array}{c} S \in \{+1, -1\} \\ S \sim \text{Uniform}\{+1, -1\} \end{array} \xrightarrow{\quad} \boxed{Y = S + Z} \xrightarrow{\quad}$$

$Z \sim N(0, 1)$

for $f_Y(y)$
to be valid
pdf $\sum p_i = 1$.

In earlier examples
you found.

$$P_{S|Y}(s|y), s \in \{+1, -1\}$$

What is $f_Y(y)$?

$$A = \{s=1\}, A^c = \{s=-1\}$$

$$f_Y(y) = P(A) f_{Y|S}(y|1) + P(A^c) f_{Y|S}(y|-1)$$

$$\begin{aligned} Y|S=1 &\sim N(1, 1) \\ Y|S=-1 &\sim N(-1, 1) \end{aligned} = \frac{1}{2} f_{Y|S}(y|1) + \frac{1}{2} f_{Y|S}(y|-1)$$
$$= \frac{1}{2} \frac{e^{-\frac{(y-1)^2}{2}}}{\sqrt{2\pi}} + \frac{1}{2} \frac{e^{-\frac{(y+1)^2}{2}}}{\sqrt{2\pi}}$$

$$M_Y(s) = E[e^{rs}]$$

$$= \int_{-\infty}^{\infty} e^{ys} f_Y(y) dy.$$

$$= \int_{-\infty}^{\infty} e^{ys} \sum_{i=1}^n p_i f_{X_i}(y) dy.$$

$$= \sum_{i=1}^n p_i \left[\int_{-\infty}^{\infty} e^{ys} f_{X_i}(y) dy \right].$$

$$= \sum_{i=1}^n p_i M_{X_i}(s).$$

Example:

IITH's SBI bank has 3 employees, two of them are slow and one fast. Time to assist is exponentially distributed with parameter $\lambda = 6$ for ^{the} fast one & $\lambda = 2$ for the slow ones.

Boltu visits the bank & picks an employee at random. What is the pdf of time it takes to assist Boltu & corresponding MGF?

$$f_{X_2}(x) = f_{X_1}(x) = 2e^{-2x} \quad x > 0.$$

$$f_{X_3}(x) = 6e^{-6x} \quad x > 0.$$

$$f_X(x) = \left[\frac{1}{3} (2 \times e^{-2x}) + \frac{1}{3} (2 \times e^{-2x}) \right]$$

↓ Boltz's wait time

$$+ \frac{1}{3} (6e^{-6x}).$$

$$M_Y(s) = \frac{2}{3} \left[\frac{2}{2-s} \right] + \frac{1}{3} \left[\frac{6}{6-s} \right].$$

$$\boxed{s < 2}$$

Joint MGF

Ref: 7.7 Sheldon M. Ross.

$$M_{\substack{(t_1, t_2, \dots, t_n) \\ X_1, X_2, \dots, X_n}} = E \left[e^{t_1 X_1 + t_2 X_2 + \dots + t_n X_n} \right]$$

$$M_{X_1}(t_1) = M_{\substack{X_1, \dots, X_n}}(t_1, 0, 0, \dots, 0).$$

- ① Inversion property: Joint MGF uniquely determines the joint distribution of $X_1, \dots, X_n$.
- ② Independence property:
- $$M_{X_1, \dots, X_n}(t_1, \dots, t_n) = \prod_{i=1}^n M_{X_i}(t_i) \quad \forall t_1, \dots, t_n \in \mathbb{R}^n$$
- iff
- $X_1, \dots, X_n$ are independent.
-

Example: @ For X, Y that are jointly Gaussian
 $(X, Y) \sim N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$.
 $\text{Cov}(X, Y) = 0 \iff X, Y \text{ are independent.}$

$$f_{X,Y}(x,y) = \frac{1}{\sqrt{2\pi} \sqrt{1-\rho^2 \sigma_x \sigma_y}} e^{-\frac{1}{2} \left[\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-\mu_y)^2}{\sigma_y^2} - 2\rho \left(\frac{x-\mu_x}{\sigma_x} \right) \left(\frac{y-\mu_y}{\sigma_y} \right) \right]}$$

If $\text{Cov}(X,Y) = \rho \sigma_x \sigma_y$.

$$\rho(X,Y) = \rho$$

If $\text{Cov}(X,Y) = 0 \implies \rho(X,Y) = 0$.

$$\implies f_{X,Y}(x,y) = \frac{1}{\sigma_x \sigma_y} \underbrace{e^{-\frac{1}{2} \left(\frac{x-\mu_x}{\sigma_x} \right)^2}}_{f_X(x)} \underbrace{e^{-\frac{1}{2} \left(\frac{y-\mu_y}{\sigma_y} \right)^2}}_{f_Y(y)}$$

$\implies X, Y$ are independent. $f_X(x) f_Y(y)$

⑥

X, Y are independent & identical.

Gaussian R.V.s then $\underbrace{X+Y}_Z, \underbrace{X-Y}_W$ are independent.

Can check $\text{Cov}(X+Y, X-Y) = 0 \Rightarrow X+Y, X-Y$ are independent

$$\begin{aligned} M_{Z,W}(t_1, t_2) &= E[e^{t_1 Z} e^{t_2 W}] \\ &= E[e^{t_1(X+Y)} e^{t_2(X-Y)}] \\ &= E[e^{(t_1+t_2)X} e^{(t_1-t_2)Y}] \\ &= M_X(t_1+t_2) M_Y(t_1-t_2). \end{aligned}$$

$$= e^{\mu(t_1+t_2)} e^{\frac{(t_1+t_2)^2 \sigma^2}{2}}$$

$$e^{\mu(t_1-t_2)} e^{\frac{\sigma^2(t_1-t_2)^2}{2}}$$

$$= e^{2\mu t_1} e^{\frac{\sigma^2}{2} \times (t_1^2 + t_2^2)}$$

$$= \boxed{e^{2\mu t_1 + \sigma^2 t_1^2}} \boxed{e^{\sigma^2 t_2^2}}$$

$\downarrow$
 fn of t_1
function of t_2 .

$\Rightarrow X+Y, X-Y$ are independent.

by independence property.